

Digital Material Passport

Manufacturer	Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	Advanced Nuclear Systems Ltd. Energy Square 456 Abingdon OX14 3DB GB procurement@advnuclear.example.com	Order	PO-56789 · 200 kg Pos. 2 · 2025-04-10
		Delivery	DN-12345 · 200 kg Pos. 1 · 2025-05-15

Product

Product Name	Stainless Steel 316L	Number (UNS)	S31603
Batch ID	H-87654-01	Name (EN)	1.4404
Surface Condition	2B	Shape	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm
Weight	200 kg	Marking	316L stamped on corner with medium depth and good legibility.
Production Date	2025-05-14	Coloring	Natural finish with full coverage for protection.
Country of Origin	DE	Product Norm	ASTM A240 (2023)

Chemical Analysis

Heat Number H-87654 - Melting Process EAF+AOD+LF - Casting Method ContinuousCasting - Casting Date 2025-05-13 - Sample Location Ladle

Symbol	C	Cr	Ni	Mo	Mn	Si	P	S	N
Unit	%	%	%	%	%	%	%	%	%
Min		16.0	10.0	2.00					
Max	0.030	18.0	14.0	3.00	2.0	0.75	0.045	0.030	0.100
Actual	0.018	17.2	10.5	2.15	1.4	0.38	0.025	0.002	0.052

Mechanical Properties

Tensile Strength ¹	Symbol	1	2	3	Average	Min	Max	Spec Min	Method	Status	
Value [MPa]	Rm	578	580	582	580	578	582	515	ASTM E8	✓	
0.2% Yield Strength ¹	Symbol	1	2	3	Average	Min	Max	Std Dev	Spec Min	Method	Status
Value [MPa]	Rp0.2	238	240	242	240	238	242	2	205	ASTM E8	✓
Elongation ¹	Symbol	1	2	3	Average	Min	Max	Spec Min	Method	Status	
Value [%]	A	51	52	53	52	51	53	40	ASTM E8	✓	

¹ 3 specimens tested

Supplementary Tests

Test	Result / limits	Method	Status
Intergranular Corrosion Resistance	Yes - No evidence of intergranular attack	ASTM A262 Practice E	✓
Pitting Corrosion Resistance 72 hours at 22°C in 6% FeCl ₃	1.2 g/m ² - max 4.0 g/m ²	ASTM G48 Method A	✓
Crevice Corrosion Resistance 72 hours in 3.5% NaCl solution	Yes - No visible crevice corrosion	ASTM G78	✓
Stress Corrosion Cracking Resistance Boiling 42% MgCl ₂ solution, 100 hours	Yes - No cracking observed	ASTM G36	✓
Ferrite Content	2.5 % - max 5.0 %	ASTM A800	✓
Grain Size	7 ASTM No. - min 5 ASTM No.	ASTM E112	✓
Inclusion Rating Worst field rating	A1, B1, C1, D1 - max A2, B2, C2, D2	ASTM E45 Method A	✓
Ultrasonic Examination	Yes - No recordable indications	ASTM A388	✓
Liquid Penetrant Examination	Yes - No relevant indications	ASTM E165	✓
Weldability Test	0.4 mm - max 1.0 mm	Varestraint Test	✓
Surface Finish	25 µin Ra - max 32 µin Ra	ASME BPE SF1	✓

PREN (Pitting Resistance Equivalent Number) Calculated using formula: $\%Cr + 3.3 \times \%Mo + 16 \times \%N$	25.8 - min 24.0		✓
Dimensional Tolerance	-0.3 - 0.2 mm - target -0.4 - 0.4 mm	ASTM A480	✓
Flatness	4 mm/m - max 9 mm/m	ASTM A480	✓
PMI (Positive Material Identification)	Yes - Material confirmed as 316L stainless steel	XRF Analysis	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM A240/A240M and meets all specified requirements. This material is suitable for nuclear applications in accordance with RCC-M code.	Validated by Thomas Wagner , Metallurgist, Quality Assurance - 2025-05-16
	Validated by Anna Schmidt , Quality Manager, Quality Assurance - 2025-05-16